

2024 / 23,262 DEVELOPERS WORLDWIDE

State of Developer Ecosystem 2024: Language Trends, Satisfaction, and Strategic Shifts

A data-driven report for technology strategy and hiring priorities across language usage, preference, and application domains.

MOST USED OVERALL

JavaScript 61%

#1 PRIMARY LANGUAGE

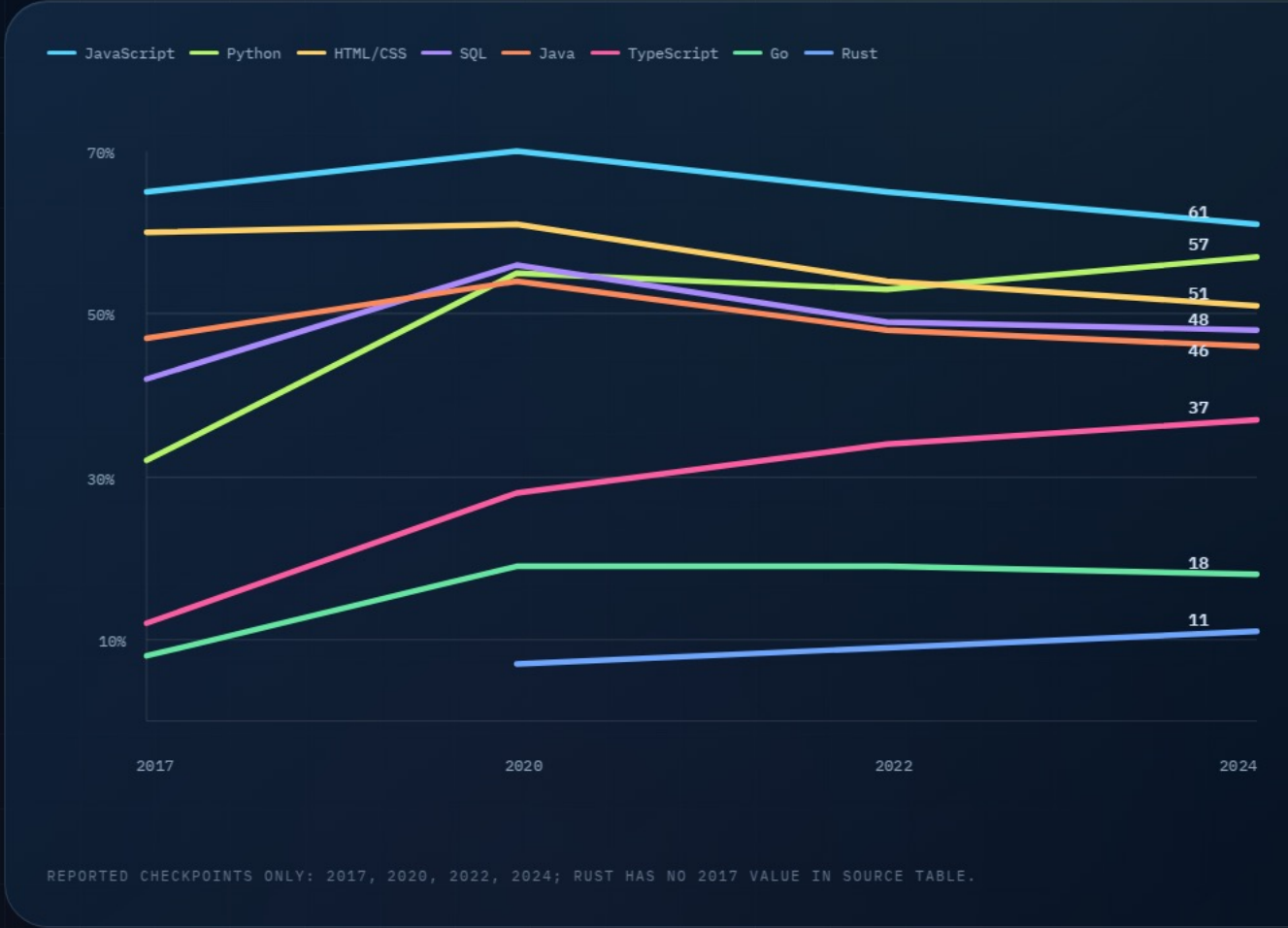
Python 35%

TOP SATISFACTION

TypeScript 199

Python Closes the Gap: 8 Years of Language Usage Shifts

JavaScript remains the most used language at 61%, but Python has risen from 32% to 57%. TypeScript tripled from 12% to 37%, while PHP declined from 30% to 17%.



Python Is Now the #1 Primary Language at 35%

Developers show strong loyalty to their chosen ecosystems. Adoption intent remains modest across all languages, suggesting expertise investment keeps teams in their comfort zone.



TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

The strongest satisfaction scores cluster around languages associated with safety, modern tooling, and expressive type systems.

TYPESCRIPT + RUST DEFINE THE DX BENCHMARK

TypeScript

Rust

Python

Go

Lua

C++

Kotlin

Shell

Dart

C



Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Language choice is increasingly domain-specific: Python leads AI and data work, TypeScript leads UI, Java owns enterprise application surfaces, and C++ remains strongest for lower-level systems.

APPLICATION DOMAIN	LEADER	SHARE	HEAT
AI / Machine Learning	Python	31%	
Data Science / Analytics	Python	28%	
Data Processing & Analytics	Python	17%	
User Interface	TypeScript	62%	
APIs & Services	Java	49%	
Application Logic & Workflows	Java	55%	
System Tools & Components	C++	27%	

ALSO REPORTED: JAVASCRIPT + TYPESCRIPT DOMINATE UI AT 58% AND 62%; C++ LEADS HARDWARE INTERFACING AT 21% AND GRAPHICS AT 15%.

What This Means for Your Technology Strategy in 2025

Translate language signals into hiring, platform, and developer experience priorities.

01

Invest in Python for AI/ML teams

Python leads AI / machine learning at 31% and data science / analytics at 28%.

02

Standardize TypeScript for UI and full-stack DX

TypeScript leads UI at 62% and tops satisfaction with a score of 199.

03

Treat Go and Rust as infrastructure growth plays

Go stabilized around 18% usage; Rust continues climbing to 11%.

04

Plan legacy web hiring carefully

PHP fell from 30% to 17%; Ruby fell from 10% to 4%.

05

Use tooling quality as a retention lever

TypeScript and Rust satisfaction reflect demand for safety and modern tooling.